MIL-P-0037232A(DM)
16 October 1985

USED IN LIEU OF
MIL-P-37232
20 May 1975

MILITARY SPECIFICATION

POINT, PULP CANAL, DENTAL, STANDARDIZED, GUTTA PERCHA

This military specification has been prepared by the Defense Personnel Support Center (DM) based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-P-37232. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers standardized (core) gutta percha pulp canal points.

1.2 Classification. The pulp canal points shall be of the size numbers indicated in Table I (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

FSC 6520

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SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging of.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B209	Sheet and Plate, Aluminum and Aluminum Alloy.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The material shall be high grade gutta percha or balata which shall be washed and treated to remove all impurities and provide a uniform, clean, non-toxic point. A filler containing barium sulfate for radiopacity shall be uniformly distributed in the finished point. All ingredients, including color additives, shall be of a high quality suitable for use in the formulation of pulp canal points.

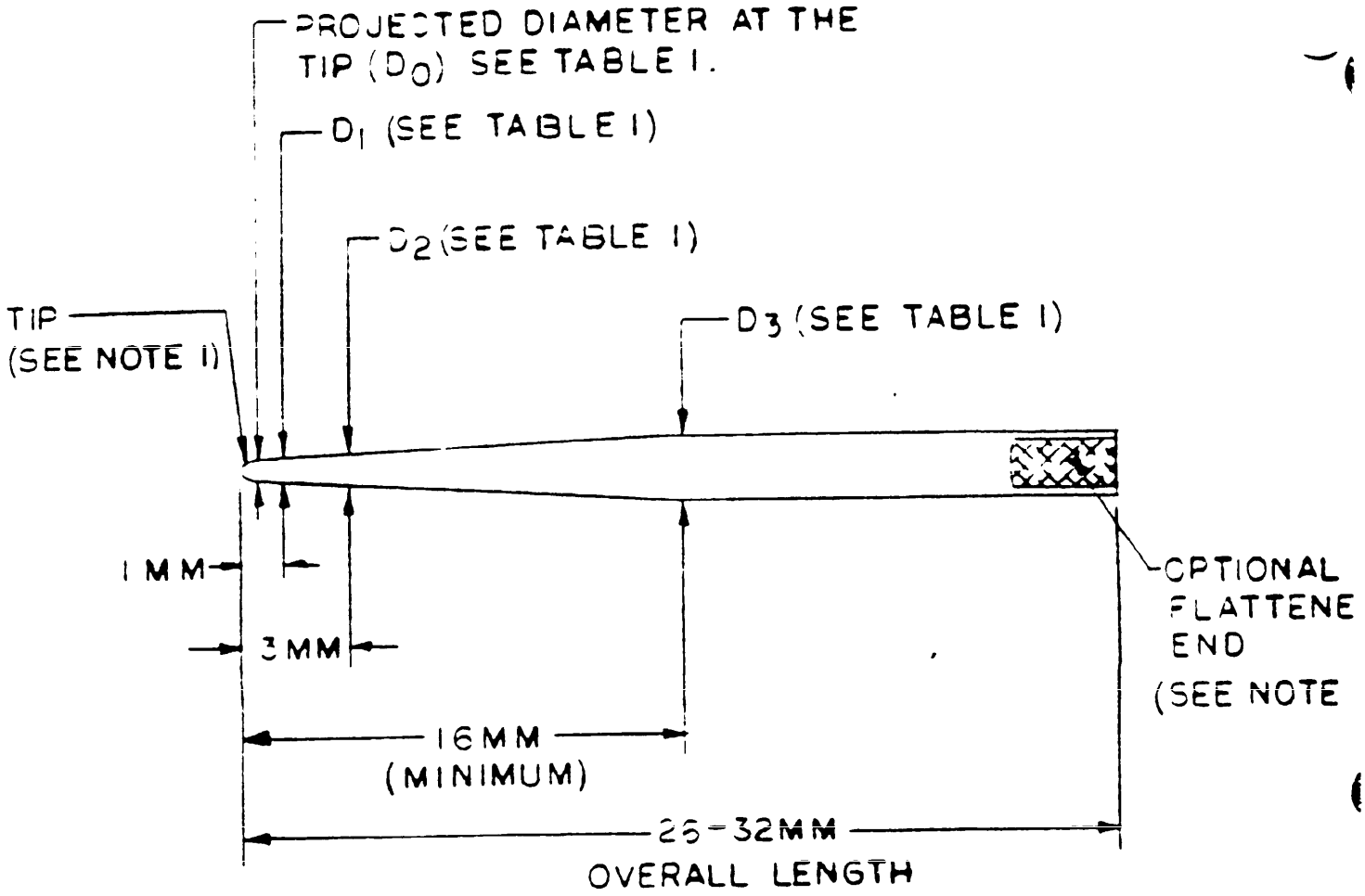
3.2 Style, design, and dimensions. The points shall be designed for use as a core point in filling pulp canals in endodontic procedures. The points shall be styled in accordance with the standardized system of the American Association of Endodontists with a standard taper of 0.02 millimeters (mm) of diameter per mm of point length from distal tip to D2 (see figure 1 and Table I). The points shall be in accordance with figure 1. The overall length of the points shall be 26-32 mm. The points shall be furnished in the sizes and corresponding diameters as shown in Table I for applicable size number designation (see 6.1). The points shall be machine rolled or hand rolled to the required dimensions.

TABLE I. Diameters of Points

Size No. (Point No.) 1/	D0 (mm) 2/	D1 (mm) 3/	D2 (mm) 4/	D3 (mm) 5/
25	0.25	0.27	0.31	0.57
30	0.30	0.32	0.36	0.62
35	0.35	0.37	0.41	0.67
40	0.40	0.42	0.46	0.72
45	0.45	0.47	0.51	0.77
50	0.50	0.52	0.56	0.82
55	0.55	0.57	0.61	0.87
60	0.60	0.62	0.66	0.92
70	0.70	0.72	0.76	1.02
80	0.80	0.82	0.86	1.12
90	0.90	0.92	0.96	1.22
100	1.00	1.02	1.06	1.32
110	1.10	1.12	1.16	1.42
120	1.20	1.22	1.26	1.52
130	1.30	1.32	1.36	1.62
140	1.40	1.42	1.46	1.72

1/ A tolerance of ±0.05 mm applies to D1, D2, and D3.
2/ D0 is the projected diameter at distal tip.
3/ D1 is the diameter 1 mm from the distal tip.
4/ D2 is the diameter 3 mm from the distal tip.
5/ D3 is the diameter 16 mm from the distal tip.

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NOTES:

1. TIP MAY BE EITHER BLUNT, CONICAL OR ROUNDED.
2. NON-OPERATIVE END: WHEN FLATTENED SHALL HAVE A SERRATED OR GROOVED SURFACE PATTERN, WHICH SHALL BE 2 - 4 MM IN LENGTH, AND AT LEAST 1/2 the D2 DIAMETER IN THICKNESS.

FIGURE 1 DIAGRAMMATIC REPRESENTATION OF STANDARDIZED CORE POINTS

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3.3 Working properties. The gutta percha points furnished shall be of such formulation and process as to provide a complete sealing of the prepared root canal when used in endodontic procedures. The working properties of the point shall exhibit cohesive strength and toughness, flexibility without brittleness.

3.3.1 Certification-Acceptability. The point shall be certified or acceptable by the American Dental Association. At time of award of contract, the manufacturer shall furnish, upon request, a certificate and data showing that the brand of material to be furnished is the same in quality and brand as that appearing on the list of acceptable or certified materials. Compliance with this certification requirement does not absolve the manufacturer from complete compliance with the other requirements of this specification.

3.4 Color. The color of the points shall be pink. The color shall be uniform throughout the point.

3.5 Radiopacity. When tested in the manner described in 4.4.2 the point material shall show a radiopacity not less than that equivalent to 4 mm of aluminum.

3.6 Quantity. Each package shall contain a minimum of 100 points of the size specified.

3.7 Delivery Schedule. No more than 2 months shall have elapsed from date of manufacture of the points to date of delivery to the Government.

3.8 Shelf Life. The points shall comply with requirements of this specification for no less than 18 months from date of delivery. When stored at 35-46 degrees fahrenheit, the inspection test date shall be 18 months from date of manufacture.

3.9 Toxicity. The manufacturer shall submit a statement that the points have no toxic effects on the normal, healthy person when used for the purpose intended.

3.10 Workmanship. Points shall be free of curled or twisted portions and shall have a smooth surface free from foreign matter and defects. The points shall show no defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of compliance. Certificates of compliance, supplied by the manufacturer of the raw materials (gutta percha or Balata and fillers), may be furnished in lieu of actual performance of such testing by the contractor. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II and III (see 4.2.1, 4.3, 4.3.1 and 4.3.2) and 4.2.2.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table II. The unit of product for examination shall be the individual size point for Major and Minor defects.

Table II. Examination Criteria

Type defect	Inspection level	AQL (Percent defective)
Major	S-4	2.5
Minor	S-1	4.0

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4.2.2 Sampling for tests. Sample size for Working Properties test shall be 6 individual size points selected at random from each lot or batch. Sample size for radiopacity test shall be 3 individual size points selected at random from each lot or batch. A lot shall be rejected if any of the samples fail the test.

4.3 Examination. The points shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with 4.2.1. The points shall be examined for defects, including but not limited to, those listed in table III.

Table III. Classification of Defects

Categories	Defects
Major	
101	Points are not clean.
102	Style and design not as specified.
103	Color of points not as specified.
104	Tapered length surface not smooth rolled finish.
105	Laminations cracks or voids present.
106	Twisted or curled sections present.
107	Size not as specified.
Minor	
151	Dimensions not as specified.

4.3.1 Dimensional examination. The diameters D1, D2 and D3 shall be measured by the use of a measuring microscope, shadowgraph, or other suitable device capable of measuring the diameters to 0.01 mm as specified in table I. The lengths shall be measured as above or by use of properly calibrated micrometers, gages or other suitable devices.

4.3.2 Visual examination. Examination for cleanliness, color, shape, uniformity and workmanship shall be made without magnification. A concavity or overhang at the operative end as a result of the machine rolling of the cylindrical stock material shall not be cause for rejection.

4.4 Tests.

4.4.1 Test Conditions. Unless otherwise stated, all tests shall be conducted at a temperature of 23° ± 2°C and 50 ± 5% relative humidity. The points shall have been pre-conditioned at this temperature and relative humidity for at least 24 hours prior to testing.

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4.4.2 Radiopacity. One millimeter of cortical bone or dentin has the radiopacity equivalent of 1 mm of 1100 Al alloy. Clearly discernable images are producible with at least 2 mm thickness differentials of either material.

The radiopacity of standardized (core) points, shall be tested by the following method.

4.4.2.1 Equipment. The radiopacity of the point material shall be determined by means of comparison of a disc of the material 10 mm in diameter with a uniform thickness of 2 mm to an aluminum step wedge having incremented thicknesses of 1 mm to 10 mm. An x-ray machine capable of producing radiation at 65 KVP and 10 mA shall be used in conjunction with radiographic film of American National Standards Speed Group D to obtain a radiograph of the test specimen and the aluminum step wedge at a focal length of 400 mm (16 inches). Inherent x-ray beam filtration is to be 2.5 mm aluminum equivalent minumum. The aluminum used for beam filtration and step wedge shall be 99% aluminum in accordance with ASTM Specification B209 (typically 1100 alloy).

4.4.2.2 Procedure. Randomly mix three (3) points for each nominal size. Heat soften and compact points from the random mix of points to a pore-free mass without segregation into a ring of 10 mm internal diameter so as to make a uniform thickness disc of 2 mm between radiolucent covers.

Position the prepared specimen in the center of the x-ray film adjacent to the aluminum step wedge. Irradiate the specimen, step wedge, and film with x-rays at 65 KVP, 10 mA at a focal length of 400 mm (16 inches) for a time of 1 second (60 impulses). After development and fixation of the film at a temperature and for time periods recommended by the film manufacturer, the film should be thoroughly washed and dried. From the processed film measure the photographic density of the radiographic image of the specimen, each of the steps of the aluminum step wedge, and the totally exposed film using the densitometer. By comparison with the image of the step wedge determine the thickness of aluminum equivalent to the specimen.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. One hundred points of one size only shall be packaged in one or more clear glass or plastic vials of suitable size. Vials shall be fitted friction caps or plug type closures, in addition, vials shall contain cotton or other suitable cushioning material to prevent damage to the points. As an alternate, points may be packaged in a compartmented plastic tray. Each compartment shall be fitted with a snug fitting polyurethane insert to prevent spilling of contents. Vials or trays shall be further packaged in a snug fitting box constructed in accordance with PPP-B-676, Type I. The box shall be wrapped and sealed in a clear cellophane or plastic wrap.

5.1.1.2 Intermediate package. Twelve unit packages containing points of one size only shall be packaged in box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for dental pulp canal points shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Fifty intermediate packages containing points of one size only shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-601, overseas type or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Fifty intermediate packages containing points of one size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for dental pulp canal points shall conform to the MIL-STD-794 requirements for this level.

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5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Levels A and B. Points packed as specified in 5.2.1 and 5.2.2, shall be unitized on pallets conforming to MIL-STD-147.

5.2.4.2 Level C. Points packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B, and C. Each unit pack, intermediate package, exterior container, and unitized load shall be marked as specified in MIL-STD-129. Type II shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall also include the lot number and legend: "STORE BETWEEN 2 -8 C (35 -46 F)" or similar statement.

NOTE: The unrefrigerated shipping time for this item shall be not more than twelve (12) days.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size Number (Point No.) required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

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NSN	Item Identification
6520-00-583-3388	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 25, 100s
6520-00-257-3030	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 30, 100s
6520-00-596-7290	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 35 100s
6520-00-257-3031	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 40, 100s
6520-00-596-7292	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 45, 100s
6520-00-257-3032	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 50, 100s
6520-00-596-7311	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 55, 100s
6520-00-257-3033	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 60, 100s
6520-00-257-3036	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 70, 100s
6520-00-257-3037	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 80, 100s
6520-00-583-3389	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 90, 100s
6520-00-583-3402	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 100, 100s
6520-01-061-6448	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 110, 100s
6520-01-061-6449	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 120, 100s

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NSN	Item Identification
6520-01-061-6450	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 130, 100s
6520-01-061-6451	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 140, 100s

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB
AGENT:
DLA-DM

Project No. 6520-1875

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYYYMMDD)*

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